Building an AI-Native Learning System for Math

Project ATLAS (AI-Native Teaching, Learning & Academic Support)

Abstract:

AI might be the highest leverage technology for education ever created. However, its promise remains unrealized by platforms that treat off-the-shelf AI products as features to plug into their existing platforms. In contrast, we propose a fundamental reimagining of data-informed individualized student support where AI-tooling and continuous experimentation underpin the entire learning-science-informed framework. Concretely, we propose building on the existing impactful *Teach to One Roadmaps* product through Project ATLAS, an initiative to develop AI-Native Teaching, Learning & Academic Support capabilities that will form the intelligence foundation of Teach to One, transforming our platform to one that is more connected, adaptive, and responsive, and capable of supporting *Roadmaps* and future learning models.

1. Introduction

K-12 math education faces a persistent paradox. We know personalized, competency-based approaches dramatically improve learning outcomes. Yet across the country, most classrooms still operate in a fundamentally one-size-fits-all model, advancing same-aged students through content by calendar rather than by demonstrated mastery. The distance between what we know works and what we practice at scale remains vast.

The consequences of this are well-documented. In 2025, TNTP and New Classrooms published *Unlocking Algebra: What the Data Tells Us About Helping Students Catch Up*, a study of approximately 2,000 students over three years (2021–2024). The findings were stark: students who lacked key "predecessor skills", specific foundational competencies from earlier grades that directly predict success in Algebra 1, had only a 13% success rate. Students with broad mathematical knowledge but missing these targeted predecessor skills achieved a 31% success rate. But students who had both broad mathematical knowledge and the critical predecessor skills reached a 58% success rate (TNTP & New Classrooms, 2025). The implication is clear: success in mathematics depends not on covering everything, but on mastering the *right* foundational skills, and doing so in a targeted, personalized way.

Artificial Intelligence (AI) can help make personalized, competency-based learning more possible at scale through student-facing supports, better teacher workflows, classroom orchestration, and richer evidence of student understanding. However, to date, AI in education has been largely defined by the chatbot. That framing has exposed real limitations: many students do not seek help effectively on their own, usage can be low, and even some of its earliest champions now see AI tutoring as only one part of a broader instructional solution. The greater promise of AI lies not in replacing teachers with conversational tools, but in strengthening the underlying systems that shape how students learn and how teachers respond. In that vision, teachers remain the relational and interpretational center of the student experience.

New Classrooms proposes a vision for that future. **Project ATLAS** builds on the foundation of *Teach to One Roadmaps*, New Classrooms' existing personalized math platform, to develop a new intelligence layer that will make *Teach to One* more connected, adaptive, and responsive, and capable of supporting additional models alongside *Roadmaps*. Many of the foundational pieces Project ATLAS will enhance, including the knowledge graph, sequencing engine, and assessment infrastructure, already exist within *Teach to One*; Project ATLAS is the next chapter of work already underway, not a new build from scratch. Project ATLAS is designed not only to enhance what New Classrooms has already built, but to create the intelligence layer and infrastructure required to ultimately support a broader set of instructional models, content providers, and school contexts.

The intelligence layer Project ATLAS will build, ATLAS, operates across multiple interconnected components, each informed by learning science research and continuously refined through empirical validation in real classrooms. The intelligence that powers a single student's personalized path is the same intelligence that helps a teacher orchestrate an entire classroom.

At the **student level**, ATLAS more deeply personalizes the progression of skills, calibrates the difficulty of challenges in real time, calibrates the difficulty of challenges in real time, detects misconceptions as they form, and dynamically responds to each learner's trajectory. At the **classroom level**, ATLAS transforms how teachers organize instruction, synthesizing learning data across students to recommend dynamic groupings, targeted modalities, and evidence-based instructional decisions that make differentiated teaching genuinely possible at scale.

This paper articulates the vision, the research foundation, the implementation pathway for Project ATLAS, and why its underlying architecture has implications not just for New Classrooms' own classrooms, but for the field. It builds on more than a decade of New Classrooms' direct work in personalized math learning, currently serving nearly 40,000 students across diverse district and school settings. That experience, including the empirical research behind *Unlocking Algebra*, has taught us what works, what doesn't, and what becomes possible when you combine sound learning science with intelligent technology and a deep commitment to equitable outcomes.

2. The Learning Science Foundation

Effective learning systems must be grounded in how students actually learn. The research points toward a coherent set of design principles that ATLAS operationalizes. Namely, students learn most from interacting with the right content, at the right time, in the right ways, for the right reasons.

The Right Content

Mastery as the Gating Criterion

While later studies have recalibrated Bloom's (1984) mastery-learning impact estimate from about 1 standard deviation[1] to .59 (Kulik et al., 1990), mastery-learning remains one of the most impactful, and simplest, effects in learning science. When students advance based on demonstrated competence rather than time elapsed, they build secure foundational knowledge that enables, rather than delays, later learning. The effect is particularly pronounced for students below grade level, for whom traditional remediation often compounds gaps rather than closing them. **The opportunity for Project ATLAS to enable per-student mastery-tracking and advancement while at the same time enabling meaningful teacher-led whole-class and small-group activities, *something no existing solution does well*.**

The Zone of Proximal Development

Vygotsky's (1978) zone of proximal development, the space between what a student can do independently and what they can do with appropriate support, remains foundational. A student who is under-challenged disengages; one who is over-challenged becomes frustrated. A personalized system can maintain each student in their zone continuously, adjusting task difficulty and scaffolding intensity in response to moment-to-moment performance. Systems that calibrate support to current competence produce measurably stronger learning gains than static scaffolding (Azevedo & Hadwin, 2005).

Predecessor Skills: The Right Foundations Matter

Which topics are in a student's zone of proximal development depends partly on what prerequisite topics they've already mastered, and not all foundational skills are created equal. The *Unlocking Algebra* research demonstrates that a specific subset of predecessor skills, competencies from earlier grades that directly enable algebraic reasoning, predict success far more powerfully than broad mathematical knowledge. The difference between a 13% and a 58% success rate in Algebra 1 was not about how much math a student had learned, but about whether they had mastered the *right* math. Effective systems must identify and target the highest-leverage predecessor skills with respect to each learning objective, pairing foundational work with opportunities to engage with new, grade-level content.

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[1] Bloom is often referenced for a "2-sigma effect," but the effect (for individual tutoring with mastery-learning features) was informally estimated from a few other works (particularly two dissertations—from Anania and Burke) which themselves involved only a few hundred students in artificial settings. Even contemporary with Bloom, studies observed mastery-learning effect sizes ranging from .6 to 1.5 standard deviations.

At the Right Time

Spaced Retrieval and Interleaved Practice

Cognitive science has established that spacing retrieval practice over time leads to dramatically stronger long-term retention (Cepeda et al., 2008), and that interleaving different problem types produces better transfer. Rohrer et al. (2014) found students who engaged in interleaved practice retained 72% of procedural knowledge one month later, compared to 38% for blocked practice. An intelligent system can orchestrate spaced and interleaved practice across an entire cohort automatically.

In the Right Ways

Formative Assessment and Real-Time Feedback

Regular, specific, actionable feedback during learning, not just summative grades at the end, produces substantively larger learning gains, with the largest effects for lower-performing students (Black & Wiliam, 1998). When students receive immediate information about what was accurate and what needs revision, they correct misconceptions before they calcify. At scale, providing this level of feedback to every student in real time requires technological support.

Active Generation

On a related note, engaging students in generating material, rather than merely reading or recognizing it, creates learning effect sizes of .4 standard deviations (Bertsch et al., 2007). This means explanations that students participate in, testing before explaining, and rich student expression formats (think free-response vs. multiple-choice) all lead to deeper and more durable learning.

Collaborative Learning

Students like talking to other people and they also learn more effectively when they're in dialogue. Meta-analyses of collaborative learning (Kyndt et al., 2013) show large effect sizes (.54 standard deviations) from engaging students in collaborative learning activities.

For the Right Reasons

Motivation and Engagement

Self-determination theory identifies three psychological needs that drive intrinsic motivation: competence, autonomy, and relatedness (Ryan & Deci, 2000). Research consistently shows these needs are satisfied by the core learning experience itself — where work is appropriately difficult, effort is recognized and progress is observable — far more than by gamification or extrinsic incentives. Engagement is a property of the system's design, not a feature bolted on.

The Unified Picture

These principles form a rich picture of how students learn mathematics. An effective system must operationalize all of them simultaneously: competency-based progression, calibration to the zone of proximal development, targeted predecessor skills, spaced and interleaved practice, responsive formative assessment, active social learning, and conditions for intrinsic motivation. Most existing systems address only a few of these. Project ATLAS is designed to embed all of them into every interaction, creating a compounding effect on learning outcomes.

3. The Technology Foundation

It seems like the whole world is talking about AI now, but it has a 75-year history and encompasses a broad assortment of individual technologies and capabilities. This section will clarify what's ripe for use in education and why, foreshadowing the system we propose in the sections following. Plainly put, a surprisingly small number of technology adoptions can power a profound reimagining of classroom math education.

The Bedrock: Continuous Experimentation

Traditionally, full-cycle feedback from implementation back to policy-makers, curriculum providers, and even directly-involved educational administration has been slow, noisy, and coarse. Full-scale standardized testing is expensive and time-consuming, so schools often only know what's working on an annual or sometimes quarterly basis, and that information rarely gets back to educational resource providers in any form to actually help them improve. In contrast, the massive growth of internet-native companies like Google, Amazon, Netflix, and others has been intrinsically driven by continuous data collection and experimentation at scale. Google transformed search by constantly testing: which links do people click on for which search terms; how long do they stay; do they come back to the same search to try a different link?

Education platforms are finally beginning to reach the scale where they can adopt this strategy as well, constantly learning about what content is working, how students are feeling, what teacher interventions are effective— but only if that learning is built into the system deliberately. Importantly, an experimentation infrastructure designed this way is inherently content-agnostic: the same framework that evaluates the effectiveness of New Classrooms' own content can evaluate content from any source, making it the foundation of a system that could power personalized learning beyond any single provider's materials.

Predicting and Discovering: Machine Learning

Machine learning algorithms (optimized statistical models) have been silently improving our lives for decades, largely invisibly. Shopping recommendations, low-traffic GPS routes, flight scheduling, photo auto-tagging, speech recognition, mail sorting, credit card fraud detection, and spam filtering are just a few of the examples that we largely take for granted. Education is finally aggregating large enough centralized datasets to leverage some of the same techniques, but for even more impactful purposes: how related are these topics; which test question gives more insight; what content most inspires curiosity?

Multi-modal Learning: Applied AI

There have been massive capability advances in applications of AI (such as Natural Language Processing, Computer Vision, Robotic Control, and Audio Synthesis) driven behind the scenes by larger datasets, faster computers, and better machine learning algorithms. Education has the opportunity to thoughtfully integrate these consumer-grade technologies in ways that transform their impact. Large language models can grade student reasoning. Computer vision models can understand pen-and-paper scratchwork. Code synthesis can produce rich manipulatives on-demand.

By themselves, these application-level technologies are unlikely to go beyond flashy demos in education. But when tested, evaluated, and quality-controlled through continuous experimentation, when informed by highly accurate models of student knowledge, curricular interrelationships, and learning trajectories, and when adopted for a purpose, rather than as an end in themselves, these emerging technologies have the opportunity to completely rewrite how classrooms interact with education technology.

4. What Project ATLAS Makes Possible for Students

Three Components and the Engagement EcoSystem

A student learning math within an effective competency-based learning system experiences three things in a continuous loop: the system decides *what* they should learn next, *how* they should learn it in the moment, and *whether* they've actually learned it. These three functions—sequencing, instruction, and assessment—are not independent:

- Preferences expressed over learning objectives suggest what they'll be attentive to in a formative learning experience.
- How a student engages during instruction reveals what assessment should probe; errors during learning can expose misconceptions that reflect the student's "boundary of mastery" at that particular moment in time.
- What a student demonstrates in assessment drives what the system sequences next.

Project ATLAS will give us the capability to organize these three functions into an integrated cycle, embedded in an engagement ecosystem that sustains motivation throughout. AI operates across all of them, but largely invisibly. There is no conversational interface at the center of the student experience. Instead, intelligence is woven into every decision the system makes, and what the system learns in one component feeds directly into the others.

Determining What Students Should Learn (The Personalized Roadmap)

Every student receives a uniquely sequenced progression of skills: not a single lock-step curriculum, but a personalized pathway tailored to their current proficiency, learning cadence, and teacher-set instructional goals.

Current state: Today, *Teach to One Roadmaps* operates on a robust set of heuristic rules leveraging more than a decade of New Classrooms' knowledge about how students learn math. These rules analyze assessment data, identify readiness patterns, consider classroom priorities, and recommend the next skill. This approach has proven effective within its design constraints.

AI opportunity: Machine learning (ML) models can take this ruleset much further with greater precision. ML can calibrate estimates of inter-topic relationships and student-topic readiness, and the models can be trained on a rich set of signals: formative assessment outcomes, engagement patterns (how long the student works, where they persist, where they disengage), time dynamics (learning velocity over weeks, forgetting curves from spaced retrieval data) and even semantic features of the content. Topic recommendations informed by these models can directly account for teacher-set priorities. Critically, the *Unlocking Algebra* research validates a key design principle: the system's sequencing engine should identify which predecessor skills are highest-leverage for each student, rather than attempting to fill every gap comprehensively. Models trained on years of learning interaction data can predict which sequences produce the strongest outcomes and which predecessor skills unlock the most future learning for each individual student.

Differentiating While They Learn (In-Skill Learning)

Once a student is working on a particular skill, the quality of the learning experience within that skill directly determines how much the student learns and how motivated they remain. In-skill learning involves students working through carefully designed resources, contextual explanations, worked examples, guided practice, and independent practice, with built-in responsiveness at every step.

Current state: Today, *Teach to One Roadmaps* guides students through a linear skill-specific formative experience, and provides static hints and fixed explanations calibrated to common errors.

AI opportunity: Here, there are multitudinous opportunities for AI to make formative learning experiences more responsive, motivating, and ultimately more impactful. LLM-authored rich manipulatives and adaptive hints can be automatically screened for baseline quality and then tested empirically against existing content. Adaptive difficulty calibration can automatically adjust problem complexity within a skill based on performance, maintaining the zone of proximal development without requiring teacher intervention. Real-time misconception detection can identify when a student has understood a procedure but lacks conceptual grounding, or vice versa, and recommend targeted support. Benchmark research has developed structured datasets of common misconceptions in middle school algebra, enabling AI systems to classify errors, flag hierarchical misconception patterns, and recommend targeted micro-lessons. This layer directly operationalizes Zone of Proximal Development research and formative assessment principles. The responsiveness of the system to each student's actual performance, moment by moment, is what creates conditions for learning and intrinsic motivation.

Capturing What They Have Learned (Multi-Modal Assessment)

Assessment is the function that closes the loop. Without it, the system is guessing at what a student knows, and differentiated learning has no signal for whether its approach is working. Assessment is what makes the entire cycle intelligent: it validates learning, surfaces misconceptions, and generates the evidence that drives every downstream decision the system makes. The quality of assessment directly determines the quality of personalization.

Current state: Today, *Teach to One Roadmaps* uses 5-question assessments that are rigorous and validated, but limited to what the system can automatically process based on predetermined answers. They do not provide rich feedback, support misconception analysis, or allow flexibility for richer modes of assessing, like showing work, explaining thinking, or leveraging pencil and paper.

AI opportunity: Two clear opportunities emerge. First, large multimodal models can score voice responses, image uploads of student work, and open-ended explanations with high reliability, capturing cognition in multiple forms. (This also enables more natural, adaptive, and open-ended assessment formats that get at student thinking more accurately, faster, and with less frustration.) Second, more sophisticated competency inference models can infer implicated knowledge states more accurately and separately track a student's conceptual depth, procedural fluency, situational application, and inter-topic contextualization, allowing the system to recommend different interventions based on the true source of a student's struggle. The continuous cycle of assessment, inference, and action is where formative assessment research is put into practice.

The User Engagement Ecosystem: A Quality of Every Interaction

The engagement ecosystem is not a separate layer; it is a dimension of the first three components. It is built into every interaction through automatically moderated difficulty, adaptive support, and context-aware encouragement. Students experience these properties as a responsive learning environment where challenges feel manageable and effort and progress is visible. This is intrinsic engagement, the kind that sustainable learning depends on.

Current state: Currently, *Teach to One Roadmaps* engagement ecosystem manifests as brief, context-aware messages at strategic moments: when a student begins, when they're frustrated, and when they've made progress.

AI opportunity: The opportunity to evolve this is significant: from disconnected commentary to history-aware intervention in the learning experience itself. In addition to acting as a signal for optimizing every other part of the system, the engagement layer also includes active intervention: dynamically adjusting difficulty, contextualizing and reframing experiences, and facilitating student-teacher connection.The engagement ecosystem operationalizes self-determination theory, where autonomy (the student navigating through their own roadmap), competence (difficulty calibrated to capability), and relatedness (the system and teacher are responsive to student needs) are woven into every moment.

The Connective Tissue: A System, Not a Collection of Features

What makes Project ATLAS so impactful is that it develops a systems approach to leveraging the power of AI. Rather than developing a collection of isolated features, the three core components of the student experience feed each other in a continuous loop. What students demonstrate in assessment improves the roadmap recommendations for what they should learn next, identifying which skills have been truly mastered and which require revisiting. Engagement signals such as time spent, pace of progress, and response patterns inform pacing decisions in the roadmap and difficulty calibration during in-skill learning. Errors detected during differentiated learning reveal misconceptions that reshape how assessment interprets student understanding. This feedback loop is what creates compounding leverage: each component's decisions improve in quality as the system learns more about the student.

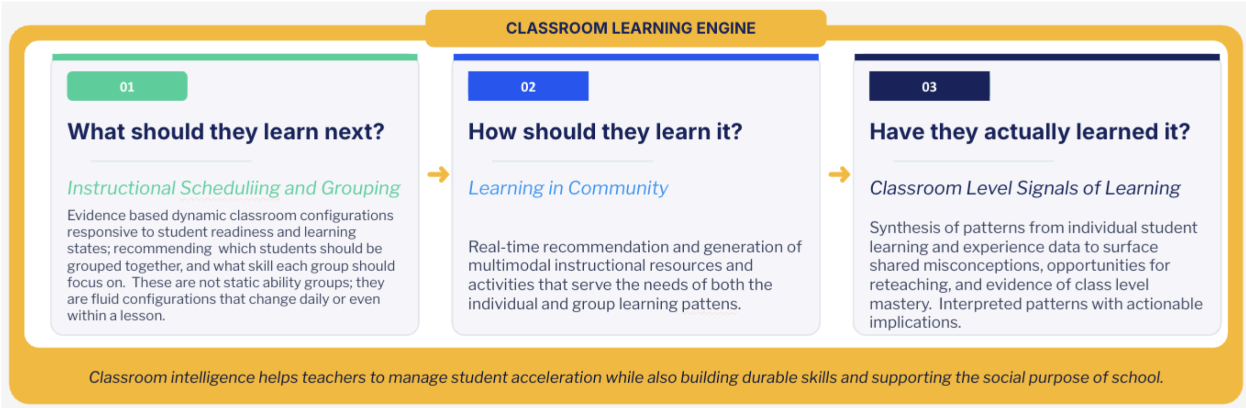
	Teach to One Current Capabilities	Powered with ATLAS
Diagnostic Assessments	Placement tests at start of year, periodic benchmarks	Continuous recalibration through every interaction, real-time proficiency mapping
In-Skill Content	Static resources, uniform difficulty across students	Adaptive hints responding to specific errors, difficulty calibrated to each student's ZPD
Skill Recommendations	Heuristic rules-based sequencing	ML models trained on longitudinal data predicting optimal paths
Skill-Level Assessments	5-question auto-scored assessments	Voice responses, pencil/paper analysis, misconception detection, open-ended explanations, and space for reflection
User Engagement	Brief, depersonified messages at key moments	Automatically moderated difficulty, adaptive hinting, context-aware encouragement, frustration reduction

5. How Project ATLAS Supports Classroom Intelligence

From Individual Pathways to Orchestrated Learning

At the individual level, ATLAS does three things: it personalizes *what* each student learns next, adapts *how* they learn it to keep them in their zone of proximal development, and validates *whether* they've actually learned it. That cycle—sequencing, adapting, validating—runs continuously for every student. But schools are not collections of isolated learners. They are classrooms, with teachers making dozens of instructional decisions every hour across students with different needs.

This is where ATLAS unlocks something powerful: **the same intelligence that drives each student's individual learning path can drive the orchestration of the entire classroom.** The system aggregates its understanding of every student, synthesizes their collective learning patterns, and surfaces instructional configurations and resources for the whole group: groupings, content focus, modalities, pacing, and materials. The teacher becomes the learning architect, and the system becomes a precision instrument that makes genuinely differentiated instruction possible at scale. This changes the teacher's work across all four of their core roles: relational, interventional, interpretational, and directional, in ways we'll explore below.



The Current and the Tide

Imagine a teacher looking at her classroom through two complementary lenses. Through one lens, she sees individual students, each with their own learning trajectory, misconceptions, and needs. This is the *current*: ATLAS powering each student's personalized path, flowing in its own direction, finding its own way. Through the other lens, she sees the collective: a classroom that shares certain readiness patterns, coalesces around particular skill clusters, and benefits from deliberate grouping and targeted instruction. This is the *tide*: the classroom-level movement that emerges when individual currents come together.

These are not separate systems. They are the same system, understood at different scales.

The **personalized roadmap**, at the classroom level, becomes an engine for **instructional scheduling and smart grouping**. The system analyzes the classroom's collective learning state, including readiness clusters, skill overlaps, and prerequisite gaps, and proposes dynamic configurations: which students should be grouped together, what skill each group should focus on, and what instructional modality that skill experience should take. These are not static ability groups that persist all year; they are fluid, evidence-based configurations that change daily or even within a lesson.

In-skill learning, at the classroom level, becomes **learning in community**. The skill experience is not only digital, it includes teacher-led mini-lessons, collaborative problem-solving with structured protocols, peer explanation pairs, mathematical discourse, hands-on manipulative work, and independent practice. Project ATLAS will provide the capability to recommend not just *what* students learn but *how*, proposing the modality and generating the instructional resources most likely to serve each group's needs based on the skill, the students' documented learning patterns, and what has worked for similar students in similar situations. Classroom experiences that wouldn't otherwise be possible are facilitated, like autonomous team formation (pre-discussion) based on which strategies students used to solve a problem, live multilingual discussions, or live synthesis of student written work.

Formative assessment, at the classroom level, becomes a source of **classroom-level signals**. Rather than individual data streams, the system synthesizes patterns across students:

shared misconceptions that warrant whole-class reteaching, evidence that yesterday's lesson landed (or didn't), and specific pedagogical recommendations for how to respond. The teacher receives interpreted patterns with actionable implications, not raw data.

The **user engagement ecosystem** at the classroom level becomes **classroom culture and collective progress**. When differentiated instruction is well-orchestrated, something shifts in the social dynamic. Students are all working on appropriately challenging problems, some at their edge of comfort, some in mastery consolidation, and they see that differentiation is normal, not a sign of deficit. Progress is visible. Growth is shared. This is intrinsic engagement at the group level, emerging from the quality of the learning experience itself.

	Teach to One Current Capabilities	→ Powered with ATLAS
Grouping & Scheduling	Teacher-created groups based on planning tool and skill-progress reports	Smart grouping: station assignments optimized across 30 personalized pathways
Learning Modalities	Static resources to support finite set of collaborative and teacher-led modalities	Modality-aware: recommends teacher-led, collaborative, digital, or independent work per group, and supports new modality creation.
Teacher Decision Support	Teacher dashboard with manual data synthesis	Real-time dashboard with signals, recommended adjustments, on-the-fly natural data synthesis & natural language trends analysis
Progress Monitoring	Progress, growth, and engagement reports, alongside individual student reports and tracking	Continuous monitoring, pattern detection across class, early warning signals, parent reporting & communication
Teacher Configurability	Teacher-inputted scope and sequence, limited ability for in-the-moment adjustments	Teacher as architect: configure pacing, grouping rules, modality preferences, and intervention thresholds

The Teacher's Role

The system proposes classroom adjustments and interventions. The teacher decides. We will be able to surface groupings and modality recommendations based on real-time data, but the teacher exercises professional judgment over all of it. She understands which student combination won't work because of social dynamics, so she regroups and the system adapts. She notices an insight emerging in student discourse and pauses to amplify it. The teacher remains the learning architect. Project ATLAS enables that architecture to be executable at scale.

In doing so, Project ATLAS will help to transform each of the teacher's four core roles. The **relational** role is enriched: with logistical burden lifted, the teacher has attention to notice what drives each student, build connections, and reflect back growth. The **interventional** role becomes targeted: the system surfaces which students need support, so the teacher focuses her expertise on designing the right intervention rather than searching for who needs it or responding only to those who self-identify as needing help. The **interpretational** role becomes more powerful: instead of drowning in raw data, the teacher receives synthesized signals with clear patterns, which she interprets through her knowledge of the student and the classroom. The **directional** role becomes connected: the teacher sets goals for individuals and the classroom that are informed by system-surfaced patterns; the system recalibrates its recommendations in service of those goals.

How Project ATLAS Enhances Professional Learning

After more than a decade of working in schools, we've learned change rarely happens overnight. Through our current work in schools, we have seen the power of in-district champions (often math coaches or academic officers) using rich classroom data to inform and direct coaching conversations, not just about implementation fidelity, but about deeper teacher practices and even content expertise. Integrating with and complementing existing workflows and expectations is important, as is providing scaffolding, professional learning, and even on-demand support as teachers grow into newly-possible practices.

Just as Project ATLAS will personalize the student experience, it will also reshape how teachers are onboarded, coached, supported in real time, and developed over the course of their careers. The same data and intelligence that personalize learning for students can make teacher onboarding adaptive to each teacher's readiness, surface in-the-moment coaching signals, give administrators real-time visibility into adoption and impact, and replace static professional learning resources with timely, context-aware support. The chart below illustrates how each dimension of teacher development shifts when current *Teach to One* capabilities are powered with ATLAS.

	Teach to One Current Capabilities	Powered with ATLAS
Teacher Onboarding	Standard training and static onboarding resources	Adaptive onboarding tailored to teacher readiness and comfort with the model and the subject matter
Administrator Support	Periodic observations and lagging implementation reports	Real-time dashboards and early signals on adoption, progress, and support needs aligned to district specific language/priorities
Real-Time Coaching	Scheduled coaching based on retrospective review	In-the-moment signals and recommendations that support live instructional decision-making
Continuous Improvement	PL effectiveness measured indirectly or infrequently	Usage and outcome data tied more directly to implementation shifts and support needs
On-Demand Support	Teachers rely on static resources or delayed help	Timely, context-aware support surfaced in response to classroom needs
Practice-Level Visibility	Limited visibility into how classroom moves affect students over time	Clearer visibility into patterns linking teacher practice, student experience, and progress

6. Beyond the Chat: Why System Intelligence is Key

The landscape of AI in education has been dominated, in recent years, by a particular vision: the chat-based AI tutor. Systems like Khanmigo promise personalized, conversational support at scale through an AI powered chat interface available to every student, responsive to their questions, adapting to their needs.

It is also, when examined closely against the research, not well-supported by evidence. A 2025 meta-analysis by Laun and Wolff reviewed studies of AI-powered chatbots in education. After correcting for publication bias (which substantially inflates reported effect sizes in educational technology research), the estimated effect on learning was small to moderate, not the large effects that early evangelists had promised. Additionally, the majority of studies documented contextual barriers to implementation: technical failures, student confusion about how to interact with the chatbots, lack of integration with curriculum, and inconsistent use. A substantial proportion of studies also note concerns about overreliance, with students asking the chatbot rather than engaging with the harder work of thinking through problems themselves (Wu et al., 2024).

By contrast, the research on adaptive learning infrastructure, such as systems that personalize sequencing, difficulty calibration, and feedback across multiple components, shows substantially stronger evidence. Wang et al. (2024) found that adaptive learning systems produce an average effect size of g = 0.70 on learning outcomes compared to non-adaptive learning interventions, translating to approximately a 26 percentile-point improvement in learner performance.

In the chatbot model, the AI-student conversation *is* the learning model. A student asks the AI a question; the AI responds, and an unstructured dialogue carries the pedagogical weight. With Project ATLAS, the learning model is the validated, competency-based system *Teach to One* already centers around: skills maps, sequenced progressions, in-skill learning, multimodal assessment, and teacher-orchestrated classrooms. AI operates *throughout* the learning model: powering the sequence engine, calibrating in-skill difficulty, scoring open-ended responses, surfacing classroom patterns. The AI is largely invisible to the student because it is doing the work that enables a smarter pedagogical experience, not trying to be the central mechanism of the learning. This is not to say that targeted AI-powered chat interfaces have no place. There are specific, bounded use cases where conversational AI can provide real value: supporting open-ended question-asking, encouraging students to elaborate on their thinking, providing after-hours support. These are targeted interventions that can be embedded within the broader system, rather than being the primary instructional tool.

7. The Ultimate Vision

Today, schools typically divide math instruction into distinct categories: core instruction, supplemental time, intervention blocks, enrichment periods, special education, and tutoring. Each operates under different rules, using different curricula, staffed by different people, and with different tools. A student might use one adaptive program in her core class and a different one during intervention. Her data does not flow between these spaces. Her learning experience is fragmented. Teachers in different roles see different pictures of the same student. Progress is siloed.

"All Math Minutes" is a vision for what becomes possible when ATLAS realizes its full capabilities to serve as the intelligent backbone of ALL of a student's math learning time, not just one block or one context. Instead of thinking about core instruction, intervention, enrichment, or tutoring as separate blocks with separate infrastructure, schools can begin to organize math around more coherent questions: *What mathematics does this student need to learn right now? What is the optimal way for them to learn it? And how will we capture what they learned?* In this future, the boundaries between instructional settings matter less than the continuity of the student's learning experience across them.

In the All Math Minutes model, rather than centering learning experiences on adherence to the broader systems of grade-level accountability, every minute of math instruction is centered around a unified understanding of the students themselves: their competencies, their gaps, their learning patterns, their goals. The system adapts its behavior based on context and setting. During a teacher-led instruction block, it informs the teacher's grouping and content decisions in real time. During independent digital time, it drives the personalized learning experience. During a practice block, it orchestrates spaced retrieval and interleaved scheduling. Every context is different, but they are all connected to the same underlying intelligence.

Teachers have a single, coherent dashboard: a unified view of each student's mathematical journey. Instead of pulling data from multiple systems and trying to synthesize a picture, a teacher sees one comprehensive roadmap: where the student is, how they got there, what comes next, and why. A special educator working with a student in a resource room sees the same roadmap the core classroom teacher sees, plus additional diagnostics on specific misconceptions. An instructional coach can see real-time data on whether the current grouping is working. Students experience genuine continuity. Their roadmap is one roadmap. Their progress is one story. Whether they are in first-period math, afternoon intervention, Saturday tutoring, or summer school, the system knows where they left off and what comes next.

School leaders gain another dimension of insight, triangulating data with other school systems to surface patterns no single source could reveal. A student who used to love math is disengaging from all of her classes and has repeated absences. A different student sees math class as a refuge, where her math teacher is best positioned to ask about her goals and dreams. A third student is excelling across the board and might need new challenges that bridge multiple disciplines.

This does not require that every minute of instruction look the same. It requires that every minute be connected: informed by the same intelligence, contributing to the same picture of the student, and part of a coherent whole.

Two Paths Forward: Digital-First and Offline-Enabled

Two parallel futures are possible as Project ATLAS matures, and they may ultimately blend.

The digital-first path is the natural evolution of the current conception of *Teach to One Roadmaps*. The platform becomes increasingly rich: more adaptive content across a wider range of math domains; deeper multimodal assessment and rich interactions (voice, image, open-ended response, handwriting); a persistent digital companion that creates continuity across sessions; and increasingly personalized recommendation engines.

The offline-enabled path challenges a fundamental assumption about where the intelligence needs to live. Many schools and families are concerned about screen time. Many students learn better with physical manipulatives and written work. Imagine a classroom where students work primarily on physical surfaces: solving problems, showing their thinking with pencil and whiteboard, collaborating with peers. A student completes a problem set, places it under a document camera, and within seconds the system has assessed it: not just whether the answer is correct, but whether her reasoning is sound, what misconceptions she might harbor, and what she's ready to learn next. The system generates her next problems, tailored to what she just demonstrated, and surfaces a synthesis for the teacher, including which peers share similar misconceptions for targeted grouping.

The component technologies exist: handwriting recognition, document scanning, AI-powered assessment, and automated content generation. ATLAS is the learning intelligence layer that connects all of them, enabling discussion, collaboration, and hands-on problem solving without sacrificing the intelligence of the system.

Whether the All Math Minutes future is primarily digital, primarily offline, or (most likely) a thoughtful blend, the system will provide the connective intelligence. The learning science doesn't change. The adaptive, competency-based, assessment-driven cycle doesn't change. What changes is the interface.

This is the long-term vision. Realizing it requires a path that respects the realities of how schools operate today, and a platform flexible enough to enter through the contexts where the need is most visible and the fit is most immediate.

8. The Path to Adoption

The All Math Minutes vision is powerful, but the K–12 market is not yet ready to adopt a fully integrated solution all at once. The constraints are structural, and they do not resolve on their own.

Policy frameworks continue to organize accountability around grade-level standards and annual, calendar-based assessments, creating persistent pressure to advance students by time rather than by mastery. Instructional mindsets across teachers, administrators, and families have been shaped by decades of grade-level pacing and still treat competency-based progression as an exception rather than a norm. Publisher ecosystems dominate core curriculum adoption on multi-year cycles, making it difficult for a new intelligence layer to displace existing materials even when the learning model underneath is stronger. And the practical realities of change management, including limited teacher capacity, finite professional learning budgets, staff turnover, and competing priorities, mean that any adoption strategy requiring wholesale transformation on day one will stall regardless of its merit.

That does not make the vision less compelling. It means the path to adoption must be sequenced, and Project ATLAS enables *Teach to One* to support that sequencing by entering schools and the broader market through contexts where the need is already visible and the fit is easier to establish.

Supplemental learning, including intervention blocks, extended-day and summer programs, enrichment periods, is the most immediate, because these contexts are already structured around individual student need rather than grade-level pacing, and *Teach to One's* adaptive, competency-based sequencing engine answers a need those programs already feel acutely.

Tutoring, particularly high-dosage and virtual models, is a similar wedge: tutors need real-time insight into what each student knows and doesn't, and the goal is acceleration rather than coverage, which is exactly what a personalized, mastery-oriented system is built to deliver.

Assessment experiences such as exit tickets and formative check-ins offer a third entry point, where *Teach to One,* powered with multimodal capabilities (voice responses, handwriting analysis, open-ended reasoning) and misconception detection, can replace a generation of multiple-choice items with something that gives teachers far more useful information.

Personalized homework is a fourth: sequencing out-of-class practice to each student's current skill level with adaptive difficulty and immediate feedback is a persistent pain point for teachers, a natural application of the system's sequencing engine, and a low-friction way to bring students and families into regular contact with it.

Core classroom adoption is also a critical opportunity. Rather than displacing a district's existing curriculum or publisher relationships, *Teach to One* can layer on top of them, using the district's scope and sequence as the frame while adding the personalization, misconception detection, and smart grouping that core materials rarely provide on their own. This path requires deeper institutional commitment than the others, but for districts already seeking better ways to differentiate within grade-level standards, it is a direct entry rather than an expansion from adjacent contexts.

And the **skill maps** themselves, the underlying knowledge graph and predecessor-skill structure that *Unlocking Algebra* validated, are directly usable artifacts that districts, publishers, and researchers can adopt as planning and analysis tools even before taking on any other part of the system.

In each case, *Teach to One* solves a concrete, visible problem while demonstrating the value of the more connected, AI-native learning system underneath. And the same flexibility that makes these entry points possible has a broader implication for how *Teach to One* enters classrooms: the platform is architected to power multiple models of instruction, not a single "mode," and to do so across more than just New Classrooms' own product.

How Project ATLAS unlocks multiple learning models

Schools organize math instruction in multiple ways, often with different pedagogical goals, staffing, and structure. *Teach to One* will be designed to integrate effectively into these multiple instructional contexts while maintaining a competency-based, personalized approach throughout. Rather than offering schools a binary choice between one product configuration and another, *Teach to One*, powered by ATLAS ,will provide a single intelligence layer that adapts to how instruction is actually structured on the ground. Three dimensions of flexibility make this possible.

Teach to One will be architected to power a **range of instructional models**, including core instruction, tutoring, enrichment, intervention, extended learning, and blended combinations of these. Each model looks different on the surface, but all of them connect to a shared adaptive intelligence and a common data foundation. A student moving across these contexts during the day will not leave *Teach to One* behind; the same sequencing engine, the same misconception models, and the same record of prior learning follow them from one context to the next.

Second, *Teach to One* is built on a modular, API-driven architecture that makes **integration possible**, streamlining connections with the systems a school or district already runs, including student information systems, rostering tools, and learning management platforms. This reduces the one-off engineering work that typically accompanies new deployments and means *Teach to One* can enter an existing technology ecosystem without wholesale replacement of the tools already in place.

Finally, traditional instructional software is organized around grade-level pacing, with the calendar as the primary structural anchor. *Teach to One* inverts this: **student progression is the organizing principle**, and personalized pathways are aligned with shared classroom goals and instructional coherence. That shift is what allows the same system to support both a grade-level core classroom where teachers are accountable to standards and an intervention block where a student may be working several grades behind, without disrupting either setting's logic.

	***Teach to One* Current Capabilities**	**Powered with ATLAS**
Model Diversity	Limited range of classroom configurations anchored in a static set of implementation models	Expands flexibility to power diverse models (core instruction, tutoring, enrichment, and intervention) all connected through a shared adaptive intelligence and data foundation.
Integration Possibilities	New partnerships and/or content sources require custom integration work, limiting speed and scalability	Modular, API-driven architecture streamlines connections with district systems, publisher content, and partner models, reducing one-off builds and enabling faster, more seamless collaboration
Organizing Principle	Grade-level pacing defines the organizing structure, limiting flexibility for individualized progress	Organizing principles center on student progression, aligning personalized pathways with shared classroom goals and instructional coherence.

Smart document processing and data import further reduce the setup burden. Bell schedules, academic calendars, pacing guides, and rosters can be ingested directly rather than manually configured, which minimizes the amount of district-specific setup work required to make each instructional context impactful. The operational ease of standing up *Teach to One* inside a new school or district is part of what makes the multi-model approach feasible at scale, not just in principle.

What this Looks Like in Practice

To make these dimensions concrete, the three vignettes below show *Teach to One*, powered by ATLAS, operating inside three different instructional contexts with three different students. The examples below illustrate how the same intelligence layer adapts to different models without changing its core value.

A Grade-Level Core Classroom

Marcus is a 6th grader in his regular math class, where his teacher, Mrs. Gonzalez, is accountable to grade-level standards and a district scope and sequence. *Teach to One* respects these constraints and personalizes within them. When Marcus begins work on expressing decimals as fractions, he starts to struggle, and the system surfaces a gap in his place-value understanding. Rather than pulling him out of the grade-level sequence, *Teach to One* orchestrates a just-in-time mini-sequence of place-value problems with hints calibrated to his exact misconception. When he treats 0.304 as 304/100, the system does not give a generic hint, it targets the error: "Is 0.304 greater than or less than 1? What about 304/100? Try thinking about the ones place and the tenths place separately." He revises, the system assesses his reasoning, and after a streak of quick answers it signals his readiness to return to grade-level fractions work. Marcus does not experience the detour as remediation. His roadmap is simply personalized.

Mrs. Gonzalez is working at a different altitude. Five minutes before class, she opens her *Teach to One* dashboard and finds five recommended groupings for the block, each with a suggested modality and curated materials, synthesized from yesterday's assessment data and this morning's entrance checks. One student is flagged for the personalized prerequisites group. She reviews the recommendation and overrides it: she knows that student has had a rough morning and will benefit from peer proximity and her direct presence, so she moves him into the teacher-led group. The dashboard updates. *Teach to One's* role is to make her decisions informed and efficient; the professional judgment remains hers.

An Intervention Block

Jayla is a 7th grader in a 30-minute intervention block that meets three times a week. She has gaps in multi-digit multiplication and fraction operations that are blocking her grade-level work in ratios and proportions. Here *Teach to One* operates with significantly wider latitude. The roadmap ranges freely across the full skill graph, working Jayla through carefully sequenced predecessor skills the system has identified as the highest-leverage for her 7th-grade readiness. The experience is rich rather than remedial, with adaptive hints, digital manipulatives, and multiple problem modalities. An interventionist monitors the session in real time and steps in when the system signals that Jayla needs human help. Nothing in the framing suggests she is "behind." Her roadmap is personalized; it simply includes deeper work on foundational skills because that is what will unlock her ability to progress.

A Summer Bridge Program

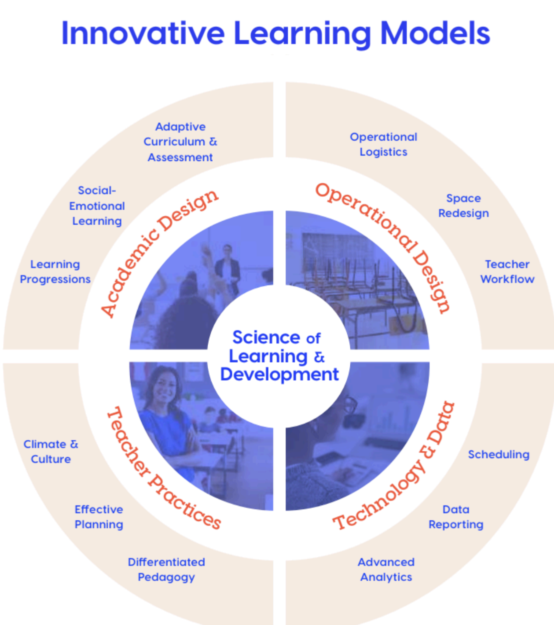
Diego is a rising 9th grader in a four-week summer bridge program aimed at Algebra 1 readiness. The program opens with a diagnostic that maps his current competencies and gaps against the Algebra 1 prerequisite graph, and *Teach to One* uses this assessment to build a four-week sequence targeting the highest-leverage predecessor skills. The summer teacher opens the dashboard and reads a synthesis in less than a minute: what Diego has mastered, what still blocks him, and what the coming weeks will work on. Across those four weeks, Diego works through carefully calibrated problems, with adaptive hints, multimodal assessment, and ongoing recalibration based on what he demonstrates. The goal is not coverage but readiness, with the *Unlocking Algebra* research grounding what readiness actually requires.

Three students, three contexts, three distinct ways *8.4*

Teach to One adapts. What changes across them is the scope of personalization, the relationship to external curriculum constraints, and the degree of freedom the system has to range across the skill graph. What stays constant is the learning science underneath: competency-based progression, predecessor-skill targeting, adaptive calibration, and continuous formative assessment.

Project ATLAS enables Infrastructure for the Field

The same flexibility that lets *Teach to One* adapt across instructional contexts within a single school also makes it usable outside New Classrooms' own classrooms. As *Teach to One* matures through Project ATLAS, the platform will be configurable for use by other organizations: curriculum providers, districts, and innovative model designers looking to embed personalized, competency-based learning into their own approaches without having to build the underlying infrastructure themselves.



In 2022, New Classrooms and Transcend published *Out of the Box: How Innovative Learning Models Can Transform K-12 Education*, arguing that the transition from the industrial paradigm of schooling to a student-centered paradigm requires a new type of organization: the innovative model provider. These are organizations that weave together instructional design, pedagogy, operations, and technology into comprehensive learning models that schools can adopt and adapt. The paper identified a critical structural barrier to this work: too few such providers exist, in large part because the technical infrastructure required, including skill maps, sequencing engines, assessment systems, experimentation pipelines, and real-time data synthesis, is extraordinarily difficult and expensive to build from scratch.

Project ATLAS has the potential to remove that barrier. The same infrastructure that powers *Teach to One Roadmaps*, including the skill map, the competency inference models, the adaptive sequencing engine, the experimentation framework, and the classroom orchestration layer, can be offered as a managed platform that other organizations build on top of. Three audience patterns illustrate how this plays out.

Curriculum providers will plug their content into *Teach to One* and immediately gain access to personalized sequencing, adaptive difficulty calibration, and multimodal assessment, without building any of that infrastructure themselves. A publisher integrating their math curriculum can continue to focus on pedagogy and content quality while *Teach to One* handles personalization underneath.

Districts will configure *Teach to One* with their own content and scope and sequence, infusing their pedagogical point of view by adjusting the calibration of modalities and the balance of inquiry-based experiences and direct instruction. A district deploying *Teach to One* across fifty schools can calibrate the system to reflect the district's instructional philosophy and local curriculum, while the intelligence layer handles personalization and orchestration underneath.

Innovative model providers, the type of organization *Out of the Box* called for, focus on what makes their model distinctive: learning design, teacher support, community relationships, and the specific experience they are building. *Teach to One* handles the computational complexity of personalization at scale. A microschool network using *Teach to One* for adaptive math while designing its own blended learning model is one example; a tutoring organization building its practice around real-time mastery signals is another.

New Classrooms has always believed that personalized, competency-based learning should be the norm, not the exception. **Project ATLAS is how that belief becomes infrastructure for the field**, evolving *Teach to One* from a platform that powers New Classrooms' own product into one that others can build on. It also reflects New Classrooms' position as a nonprofit: the organization's mission is to shift how K-12 math is taught and learned at scale, not to capture a single slice of the market. A platform approach lets New Classrooms partner with organizations that might otherwise be in competition with each other, and with organizations that could not reasonably fund this infrastructure on their own.

The Shared Foundation

Across every instructional context *Teach to One* supports and every platform configuration it enables, the underlying learning science stays constant. Competency-based progression, targeted predecessor skill development, adaptive difficulty calibrated to each learner's zone of proximal development, continuous formative assessment, and the compounding intelligence of the ATLAS-powered learning cycle are not features that apply in some settings and not others. They are how the system works, full stop.

What changes from one model to the next is the scope of personalization, the relationship to external curriculum constraints, and the degree of freedom the system has to range across the skill graph. A grade-level classroom constrains the roadmap to on-grade standards while still differentiating within them. An intervention block opens the roadmap to the full skill graph so the system can target predecessor skills wherever they sit. A summer program prioritizes readiness for the next grade. A district configuration reflects that district's pedagogical point of view. A publisher integration routes through the publisher's content. In each case the configuration is different, but the learning principles and the accumulated record of the student are the same.

This is what makes *Teach to One* a platform with intelligence at its core, rather than a static piece of software. The system's behavior adapts to context, but what it does for a given student, meaning measuring what they know, identifying what they need next, calibrating the experience to their current zone, and closing the loop with evidence of learning, does not. That is the claim New Classrooms is in a position to make.

9. New Classrooms Is Uniquely Positioned

Project ATLAS builds on more than a decade of New Classrooms' direct work in personalized math learning, including the development and scaling of *Teach to One Roadmaps*, its existing product serving nearly 40,000 students across diverse district and school settings. This is not a vision starting from zero. It is the next chapter of work already underway: extending the learning science, product infrastructure, and classroom experience developed through *Teach to One Roadmaps* into a more deeply AI-native foundation for teaching, learning, and academic support.

New Classrooms has invested more than $100 million over the fifteen years building the technical infrastructure and organizational know-how required to support personalized, competency-based learning. That investment has produced assets that are difficult to replicate: years of historical implementation data, a proprietary mathematics skill map, a structured dataset of more than 5 million student-skill interactions generated within a personalized, competency-based learning framework, and a platform architecture already operating in real classrooms. The *Unlocking Algebra* research, conducted on New Classrooms' own student population data in partnership with TNTP, further demonstrates that this foundation is not merely structural but empirically validated: the predecessor skill relationships in the system predict real-world student success.

Much of what Project ATLAS requires already exists in foundational form. The skill maps, competency inference models, adaptive sequencing engine, experimentation infrastructure, data pipelines, content systems, and classroom orchestration layer are not hypothetical. They are already in use through *Teach to One Roadmaps* and can now be enhanced through AI and tested in authentic school settings. This gives New Classrooms a rare ability to apply emerging AI capabilities to longstanding instructional and operational challenges in ways that are grounded in practice, not built in abstraction.

The team is built for this work. It includes academicians, product leaders, technologists, computer science PhDs, and former teachers who understand both the possibilities of adaptive systems and the realities of classroom implementation. New Classrooms' approach is grounded in learning science, not hype. Personalized, competency-based learning is not a marketing frame for the organization; it has been the core of its work for fifteen years.

Moreover, as a nonprofit, New Classrooms is positioned to pursue Project ATLAS not only as product innovation, but as infrastructure for the broader field. It is focused on a mission of transforming K–12 education, not on generating financial returns for investors. That creates unusual freedom to partner across the sector, including with organizations that may otherwise compete with one another or lack the resources to build the necessary infrastructure and data on their own.

To realize the full ATLAS vision, both as an enhancement to *Teach to One* and as infrastructure for the broader field, additional capabilities will need to be layered in: diverse content ingestion so partner curricula can be integrated alongside New Classrooms' own content; configurable scope and sequence constraints so *Teach to One* can operate across different pedagogical frameworks; partner-facing APIs and interfaces; and a multi-tenant architecture that enables multiple organizations to build on the same intelligence layer while maintaining their own identity and instructional approach.

Project ATLAS is the next chapter of this work: the application of powerful new AI capabilities to problems New Classrooms has been solving methodically for years, and a path toward making the underlying capabilities of personalized, competency-based learning more powerful and more broadly available. New Classrooms has always believed personalized, competency-based learning should be the norm, not the exception. Project ATLAS creates an opportunity not only to deepen that work within *Teach to One*, but to extend its impact across a far broader set of students, classrooms, and organizations.

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